



Samsung's NL22B transparent display

ployed 107cm (42-inch), touch-enabled HP digital signage displays and HP TouchSmart All-in-One PCs. Apart from several digital engagements, including a guess-the-spice game, it has setup an 8.5m (28-foot) wall dedicated to broadcasting McCormick's past and history of spices and flavours in an immersive experience.

J2 Retail Systems, makers of special digital displays for the retail world, also has some interesting products such as the TFR and Pro-Cap. Their True Flat Resistive (TFR) screen has a bezel-free design that makes it a totally flat, stand-alone screen that is both practical and stylish. The TFR screen's IP rating denotes that it can be used outdoors.

For exceptional durability, J2 is also equipping some of its models with Projective Capacitive Touch (Pro-Cap) screens, which apparently come with an almost unlimited touch life. These scratch-resistant glass screens are said to be durable, while also delivering stable performance with exceptional screen clarity and brightness. Pro-Cap screens also allow finger touch inputs such as gesturing, and are unaffected by moisture, temperature or harsh cleaning products. J2 offers touch screens in 5-wire resistive, surface acoustic wave (SAW) and infrared types.

Samsung has created much more excitement with three new technologies unveiled at the GITEX Technology Week held in October this year at the Dubai World Trade Centre: a transparent display (NL22B), a 190cm (75-inch)

LED LFD (ME75B) and a 3D video wall.

The NL22B transparent display can be used to create an immersive shopping experience. It may also be used in other fields such as healthcare and education. The 56cm (22-inch) transparent display is a cube-shaped device that contains a transparent front panel with a com-

puter built into the chassis. It allows images and animations to interact with the products showcased inside the cube. So it can be used in a store to allow users to interactively find out more about products, try out various combinations, etc, before making a purchase.

The ME75B is a cutting-edge, huge light-emitting diode (LED) display that is energy-efficient, smart and network-enabled. It can adjust brightness by itself, and has a built-in media player too, which makes it a standalone device. It can be used with a touch overlay and can be implemented at stores or restaurants to provide customers with an interactive guide.

The 3D video wall is a slim, edge-lit, high-definition LED panel—another product that will be loved by retailers!

Performance matters—everywhere

While touch has been throwing its weight all over the display scene, other factors such as resolution and display performance also continue being points of innovation. The issue of performance, for instance, is being dealt with from varied angles—right from the material to the chip and the software.

One interesting example is the Corning Lotus Glass technology unveiled last year. The eco-friendly, high-performance display glass has been developed to support the demanding manufacturing processes of both organic LED displays and LCDs for high-performance, portable devices such as

smart phones, tablets and notebook computers. It performs exceptionally well in low-temperature poly-silicon (LTPS) and oxide TFT backplane manufacturing environments.

"Corning Lotus Glass has a high annealing point that delivers the thermal and dimensional stability our customers require to produce high-performance displays," said Andrew Filson, worldwide commercial director, Display Technologies, and vice president, Corning Holding Japan GK, in a press report. "Because of its intrinsic stability, it can withstand the thermal cycles of customer processing better than conventional LCD glass substrates. This enables tighter design rules in advanced backplanes for higher resolution and faster response time."

To keep pace with the demand for performance, this year 3M also announced a few high-performance multi-touch displays that extend tablet-like responsiveness to larger, interactive displays. The 61cm (24-inch) M2467PW and the 69cm (27-inch) M2767PW feature brilliant high-definition graphics powered by LED backlights, wide viewing angles and elegant smart phone-like product design. The M2467PW can recognise up to 20 simultaneous touch events. The M2767PW with ultra-fast 40-touch capability is ideal for multi-user applications such as digital signage and product selectors.

Based on 3M's Projected Capacitive Technology (3M PCT), these multi-touch displays offer the unique ability to recognise and reject the user's palms or arms resting on the touch screen to help prevent inadvertent touches while also maximising multi-touch functionality. A 3M-proprietary anti-stiction surface treatment allows the user's fingers to effortlessly glide across the screen for smoother gesturing and accurate draw capabilities.

Microsoft never stands out of any race! This March, researchers at Microsoft claimed to have developed a touch screen display that improves touch screen latency by a factor of 100. Current touch screen displays have an input latency of about 100 ms, which